Technical Appendix for scd Package (Draft)

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This appendix establishes the mathmatical equivalence between the theoretical estimators defined in Section 5 of the main text and the computational implementation provided in the scd package. While the theory is presented using standard algebraic definitions, the software employs specific matrix decompositions and algorithmic optimizations to ensure numerical stability and to accommodate survey weights. We detail these mappings below.

1 Weight Estimator

1.1 Theoretical Object

The Synthetic Control with Differencing (SCD) weights $\hat{w}$ are defined in the main text (Equation 5.4) as the minimizer of the squared pre-treatment matching errors:

$$\hat{w} = \underset{w \in \Delta_{K-1}}{\arg\min} \sum_{t \in \mathcal{T}_0} \left(\hat{\mu}_{0,t} - \sum_{j=1}^{K} \hat{\mu}_{j,t} w_j \right)^2 . \tag{1}$$

1.2 Computational Implementation

The script `att_scd.R` generalizes the estimator to support survey weights π_i. The inputs $\hat{\mu}_{j,t}$ are computed as weighted means:

$$\tilde{m}_{j,t} = \frac{\sum_{i=1}^{n} \pi_i Y_i \mathbb{I}(G_i = j, T_i = t)}{\sum_{i=1}^{n} \pi_i \mathbb{I}(G_i = j, T_i = t)} = \frac{\sum_{i \in \text{cell}_{j,t}} \pi_i Y_i}{\tilde{n}_{j,t}}, \tag{2}$$

where $\tilde{n}_{j,t} = \sum_{i \in \text{cell}_{j,t}} \pi_i$ is the sum of weights for the group-time cell.

In the code:

```
# Weighted group means
means_ordered <- collap(data, Y ~ t + G, w = ~ weight)
```

The vector $\boldsymbol{y}_{\text{pre}}$ (code: `hat_gamma_pre`) and matrix $\mathbf{X}_{\text{pre}}$ (code: `hat_Gamma_pre`) are constructed from these weighted means. The objective function is implemented as:

```
diff_vec <- hat_gamma_pre - hat_Gamma_pre %*% w
objective = crossprod(diff_vec)
```

The `crossprod` function computes the inner product, expanding to $\sum_{t=1}^{T_0} (\hat{\mu}_{0,t} - \sum_{j=1}^{K} \hat{\mu}_{j,t} w_j)^2$. The constraints $w \geq 0$ and $\sum w_j = 1$ are enforced via the `nloptr` optimization routine.

2 Asymptotic Variance Matrix $\hat{V}_{RC}(w)$

2.1 Theoretical Object

Equation (5.10) defines the variance matrix as:

$$\hat{V}_{RC}(w) = \frac{1}{T^* - 1} \sum_{t=1}^{T^*-1} \left[\sigma_t^2(w) \times B_2' \mathbf{M}_t B_2 \right], \tag{3}$$

where $\sigma_t^2(w)$ is the scalar variance of the weighted residuals at time t, and $\mathbf{M}_t$ is the outer product matrix of the centered means:

$$\mathbf{M}_t = \frac{\hat{\boldsymbol{\mu}}_t \hat{\boldsymbol{\mu}}_t'}{T^* - 1} - \lambda_t(\bar{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}_t' + \hat{\boldsymbol{\mu}}_t\bar{\boldsymbol{\mu}}') + (T^* - 1)\lambda_t^2 \bar{\boldsymbol{\mu}}\bar{\boldsymbol{\mu}}'. \tag{4}$$

Here, B_2 is the $K \times (K-1)$ matrix of eigenvectors corresponding to the non-zero eigenvalues of the centering matrix $I_K - \frac{1}{K}\mathbf{1}\mathbf{1}'$.

2.2 Computational Implementation

The script `gen_hat_V.R` computes this sum via an accumulation loop that explicitly accounts for survey weights.

Step 1: The Scalar Variance $\sigma_t^2(w)$. The asymptotic variance of the weighted mean involves the sum of squared weights. Inside the loop over t, the variable `n_obs` is computed as $\sum_{i \in \text{cell}} \pi_i^2$. The code executes:

```
# Influence function scaled by n / sum(weights)
hat_psi_it_star[G_i, 1] <- (n / n_jt_t[G_i]) * (Y_i - group_means_t[G_i])

# Inner difference term
inner <- (drop(t(hat_psi_it_star[2:(K + 1), drop = FALSE]) %*% w) - hat_
    psi_it_star[1, 1])^2

# Accumulate contribution weighted by Sum(pi^2)
hat_V_it <- hat_V_it + n_obs_sq * inner
```

Here, the variable `hat_psi_it_star` corresponds to the weighted influence function:

$$\hat{\psi}_{ij,t}^* = \frac{n}{\tilde{n}_{j,t}}(Y_i - \tilde{m}_{j,t}).$$

The accumulation term `n_obs_sq * inner` effectively computes:

$$\sum_{i \in \text{cell}} \pi_i^2 \left(\hat{\psi}_{ij,t}^* \right)^2.$$

Dividing by n (as done at the end of the loop) yields the correct asymptotic variance for ratio estimators in repeated cross-sections. In the unweighted case ($\pi_i = 1$), `n_obs_sq` reduces to the cell count and $\tilde{n}_{j,t}$ to the number of observations, recovering the standard formula.

Step 2: The Matrix Component $\mathbf{M}_t$. The script constructs the matrix $\mathbf{M}_t$ (variable `M`) directly from the vector of means at time t (`mu_t`) and the grand mean (`bar_mu`):

```
M <- (1 / (T_star - 1)) * (mu_t %*% t(mu_t)) -
     lambda[pre_t] * (bar_mu %*% t(mu_t) + mu_t %*% t(bar_mu)) +
     (T_star - 1) * (lambda[pre_t]^2) * (bar_mu %*% t(bar_mu))
```

This code block is a literal translation of the algebraic definition of $\mathbf{M}_t$.

Step 3: B_2 Transformation and Summation. The function returns the transformed term `hat_V_it * t(B2) %*% M %*% B2` for each t. The final matrix is obtained by summing these terms and normalizing by $1/(T^* - 1)$.

3 Asymptotic Variance Matrix $\hat{V}(w)$ for Balanced Panel Data

When the data forms a balanced panel, the observations are no longer independent across time. This induces cross-temporal covariance that must be accounted for in the variance estimator. We outlined the panel variance formula and establish its computational equivalence.

3.1 Theoretical Object

In the balanced panel setting, we have $N_t = N$ for all $t \in \mathcal{T}$ and $G_{i,t} = G_i$ (group membership is time-invariant). The variance matrix is defined as:

$$\hat{V}(w) = B_2' \left(\frac{1}{n} \sum_{i \in N} \pi_i^2 \hat{\boldsymbol{z}}_i(w) \hat{\boldsymbol{z}}_i'(w) \right) B_2, \tag{5}$$

where the influence vector $\boldsymbol{z}_i(w)$ aggregates information across all pre-treatment periods:

$$\boldsymbol{z}_i(w) = \alpha_{G_i} \boldsymbol{z}_{i,G_i}, \quad \text{with} \quad \alpha_j = \begin{cases} -1 & \text{if } j = 0 \text{ (treated)} \\ w_j & \text{if } j \geq 1 \text{ (donor)} \end{cases} \tag{6}$$

and

$$\boldsymbol{z}_{i,G_i} = \frac{1}{T_0} \sum_{t=1}^{T_0} \boldsymbol{\mu}_t \psi_{i,G_i,t}. \tag{7}$$

The λ-differenced influence function is:

$$\psi_{i,G_i,t} = \psi^*_{i,G_i,t} - \sum_{s=1}^{T_0} \lambda_s \psi^*_{i,G_i,s}, \tag{8}$$

where the raw influence function for individual i in group j at time t is:

$$\psi^*_{i,j,t} = \frac{n}{n_{j,t}} \mathbf{1}\{G_i = j\}(Y_{i,t} - m_{j,t}). \tag{9}$$

3.2 Computational Implementation

The script `gen_hat_V.R` implements the panel variance via the function `gen_hat_V_panel_fast`, which precomputes all individual-level influence functions once.

Step 1: Precompute Individual Influence Functions. For each individual i, the code computes the λ-differenced influence function at each period:

```
# For each individual i in group j
psi_star_i <- (n / n_jt[j, ]) * (Y_i - group_means[j, ])  # T_0-vector
psi_diff_i <- psi_star_i - sum(lambda * psi_star_i)       # lambda-
    differenced
```

Step 2: Compute $z_i(w)$ Vector. The script aggregates across pre-treatment periods:

```
# z_ij = (1/T_0) * sum_t mu_t * psi_ij_t
z_ij <- (1/T_0) * crossprod(hat_Gamma_pre, psi_diff_i)  # K-vector
z_i <- alpha[G_i] * z_ij                                 # Weighted by
    alpha
```

Step 3: Accumulate Outer Products. The variance is computed by summing weighted outer products:

```
# Accumulate (1/n) * sum_i pi_i^2 * z_i * z_i'
Sigma_z <- Sigma_z + weight_sq_i * tcrossprod(z_i)
```

Step 4: Transform via B_2. The final variance matrix is:

```
hat_V <- (1/n) * crossprod(B2, Sigma_z) %*% B2
```

This exactly implements Equation (5).

4 ATT Variance

4.1 Theoretical Object

The variance of the ATT estimator, $\hat{\theta}_t(\lambda, \hat{w})$, depends on the differencing parameter λ. Under the assumption of independent repeated cross-sections, the theoretical variance is:

$$\mathrm{Var}(\hat{\theta}_t) = \mathrm{Var}(Z_t) + \sum_{s \in \mathcal{T}_0} \lambda_s^2 \mathrm{Var}(Z_s) - 2\mathbb{I}(t \in \mathcal{T}_0)\lambda_t \mathrm{Var}(Z_t), \tag{10}$$

where $Z_r = \hat{m}_{0,r} - \sum_{j=1}^{K} w_j \hat{m}_{j,r}$ represents the prediction error at time r.

4.2 Computational Implementation

The script `gen_hat_sigma_squared.R` implements a single-pass accumulator algorithm that achieves the complexity to linear time $O(T)$.

4.2.1 Step 1: The `base2` Accumulation

The script iterates through the data once. For each time period r, it calculates the variance of the prediction error, denoted in the script as `sum_base2_per_r`.

Inside the loop over group-time cells, the script calculates:

```
# Influence function for the prediction error (Y - synthetic_Y)
s <- (n / n_jt_val) * (Y_val - m_g_r)
base <- if (G == 1) s else -w[G-1] * s
base2 <- base^2
```

Here, `base` corresponds to the influence function value $\psi^{Z}_{ij,r}$ for the variable Z_r. The term `base2` is its square.

The script then updates two global accumulators using the survey weights (`n_obs_sq` $= \sum \pi_i^2$):

```
# 1. Store the raw variance for period r
sum_base2_per_r[r] <- sum_base2_per_r[r] + n_obs_sq * base2

# 2. Accumulate the global constant (Sum of lambda^2 * Var)
total_const <- total_const + n_obs_sq * base2 * dr2
```

At the end of this loop:

- `sum_base2_per_r[r]` holds $n \times \widehat{\mathrm{Var}}(Z_r)$.
- `total_const` holds $n \times \sum_{s=1}^{T_{total}} \delta_s^2 \widehat{\mathrm{Var}}(Z_s) = n \times \sum_{s \in \mathcal{T}_0} \lambda_s^2 \widehat{\mathrm{Var}}(Z_s)$.

4.2.2 Step 2: Constant-Time Variance Retrieval

Once the loop is complete, the variance for any specific period t is retrieved in $O(1)$ time using the algebraic identity derived below. The script executes:

```
extra <- sum_base2_per_r[t] * (1 - 2 * delta[t])
hat_sigma_squared[t] <- (1/n) * (total_const + extra)
```

Mathematical Equivalence Proof: Substituting the definitions into the code's formula:

$$\hat{\sigma}_t^2 = \sum_{s \in \mathcal{T}_0} \lambda_s^2 \mathrm{Var}(Z_s) + \mathrm{Var}(Z_t)(1 - 2\delta_t).$$

Case 1: Post-Treatment ($t > T_0$)
Here, $\delta_t = 0$. The formula becomes:

$$\hat{\sigma}_t^2 = \underbrace{\sum_{s \in \mathcal{T}_0} \lambda_s^2 \mathrm{Var}(Z_s)}_{\text{Variance of Pre-trend}} + \underbrace{\mathrm{Var}(Z_t)}_{\text{Variance of Post-outcome}}.$$

This matches the variance of $\hat{\theta}_t = Z_t - \sum \lambda_s Z_s$ under independence.

Case 2: Pre-Treatment ($t \leq T_0$)
Here, $\delta_t = \lambda_t$. The formula becomes:

$$\hat{\sigma}_t^2 = \sum_{s \in \mathcal{T}_0} \lambda_s^2 \mathrm{Var}(Z_s) + \mathrm{Var}(Z_t)(1 - 2\lambda_t).$$

Expanding the summation by separating the term where $s = t$:

$$\hat{\sigma}_t^2 = \sum_{s \neq t} \lambda_s^2 \mathrm{Var}(Z_s) + \lambda_t^2 \mathrm{Var}(Z_t) + \mathrm{Var}(Z_t) - 2\lambda_t \mathrm{Var}(Z_t).$$

Factoring the terms for Z_t:

$$\hat{\sigma}_t^2 = \sum_{s \neq t} \lambda_s^2 \mathrm{Var}(Z_s) + \mathrm{Var}(Z_t)(\lambda_t^2 - 2\lambda_t + 1) = \sum_{s \neq t} \lambda_s^2 \mathrm{Var}(Z_s) + \mathrm{Var}(Z_t)(1 - \lambda_t)^2.$$

This precisely matches the variance of the pre-treatment residual $\hat{\theta}_t = (1 - \lambda_t)Z_t - \sum_{s \neq t} \lambda_s Z_s$.

This algorithmic structure ensures that the computationally intensive summation is performed only once (`total_const`), allowing the method to scale efficiently to large T without sacrificing exactness.

4.3 Panel ATT Variance: Cross-Temporal Covariance

For balanced panel data, the ATT variance $\hat{\sigma}_t^2(w)$ must account for within-individual correlation across time periods. The theoretical variance is:

$$\hat{\sigma}_t^2(w) = \frac{1}{n} \sum_{i \in N} \pi_i^2 \psi_{i,t,\theta}^2(w), \tag{11}$$

where the ATT influence function is:

$$\psi_{i,t,\theta}(w) = \psi_{i,0,t} - \sum_{j=1}^{K} w_j \psi_{i,j,t} = \beta_{G_i} \psi_{i,G_i,t}, \tag{12}$$

with $\beta_0 = 1$ (treated) and $\beta_j = -w_j$ (donor j).

Substituting the λ-differencing:

$$\begin{aligned} \psi_{i,t,\theta}^2 &= \beta_{G_i}^2 \left(\psi_{i,G_i,t}^* - \sum_{s=1}^{T_0} \lambda_s \psi_{i,G_i,s}^* \right)^2 \\ &= \beta_{G_i}^2 \left[(\psi_{i,G_i,t}^*)^2 - 2\psi_{i,G_i,t}^* \sum_s \lambda_s \psi_{i,G_i,s}^* + \left(\sum_s \lambda_s \psi_{i,G_i,s}^* \right)^2 \right]. \end{aligned} \tag{13}$$

The cross-temporal products $\psi_{i,j,t}^* \psi_{i,j,s}^*$ are non-zero for $t \neq s$ in panel data, requiring the full covariance structure.

Computational Implementation The script `gen_hat_V.R` precomputes all influence functions for each individual and then evaluates the full quadratic form:

```
# For each individual i
psi_star_vec <- (n / n_jt[G_i, ]) * (Y_vec_i - group_means[G_i, ])
psi_theta_t <- psi_star_vec[t] - sum(lambda * psi_star_vec[1:T_0])
sigma_sq[t] <- sigma_sq[t] + weight_sq_i * beta[G_i]^2 * psi_theta_t^2
```

5 Bonferroni Method

5.1 Theoretical Optimization Problem

The precomputed matrix for the Bonferroni test is:

$$P = B_2 \hat{V}^{-1} B_2', \tag{14}$$

where $\hat{V}$ is $(K-1)\times(K-1)$ and invertible, but P is $K\times K$ with $\text{rank}(P)=K-1$.

The test statistic $T_{\mathsf{bf}}(w)$ requires solving:

$$\hat{r}(w) = \underset{r\geq 0,\, w'r=0}{\arg\min}\ (\varphi - r)'P(\varphi - r), \tag{15}$$

where $\varphi = \hat{H}w - \hat{h}$.

5.2 Computational Implementation

We use OSQP (Operator Splitting Quadratic Program), which natively handles positive semi-definite Hessians. The optimization (15) is reformulated in OSQP's standard form:

$$\min_r \frac{1}{2} r'(2P)r + (-2P\varphi)'r \quad \text{subject to} \quad l \leq Ar \leq u, \tag{16}$$

where the constraint matrix encodes both $w'r = 0$ (equality) and $r \geq 0$ (inequality):

$$A = \begin{pmatrix} w' \\ I_K \end{pmatrix}, \quad l = \begin{pmatrix} 0 \\ \mathbf{0}_K \end{pmatrix}, \quad u = \begin{pmatrix} 0 \\ \infty \cdot \mathbf{1}_K \end{pmatrix}. \tag{17}$$

The implementation:

```
# OSQP formulation (handles PSD matrices natively)
P_osqp <- 2 * precomp  # precomp = B2 %*% solve(hat_V) %*% t(B2)
q_osqp <- -2 * as.vector(precomp %*% phi_diff)

A_osqp <- rbind(w, diag(K))
l_osqp <- c(0, rep(0, K))
u_osqp <- c(0, rep(Inf, K))

# Solve with OSQP
model <- osqp::osqp(P = P_osqp, q = q_osqp, A = A_osqp,
                    l = l_osqp, u = u_osqp)
result <- model$Solve()
hat_r <- result$x
```

The test statistic is computed using the optimal $\hat{r}$:

```
diff_vec <- phi_diff - hat_r
T_stat <- as.numeric(n * t(diff_vec) %*% precomp %*% diff_vec)
```

The critical value $\hat{c}_{1-\kappa}$ depends on the degrees of freedom $\hat{k}(w) = \max(K-1-\hat{d}(w), 1)$, where $\hat{d}(w)$ counts binding constraints:

```
gamma_vec <- precomp %*% diff_vec
hat_d <- sum(abs(gamma_vec) < tolerance & w == 0)
hat_k <- max(K - 1 - hat_d, 1)
hat_c_1_kappa <- qchisq(1 - kappa, df = hat_k)
return(T_stat <= hat_c_1_kappa)
```

Once the confidence set for weights $\tilde{C}_{1-\kappa}$ is determined, the CI for θ_t is:

$$c_L = \min_{w\in\tilde{C}_{1-\kappa}} \hat{\theta}_t(w) - z_{1-\beta}\cdot\hat{\sigma}_t/\sqrt{n}, \tag{18}$$

$$c_U = \max_{w\in\tilde{C}_{1-\kappa}} \hat{\theta}_t(w) + z_{1-\beta}\cdot\hat{\sigma}_t/\sqrt{n}, \tag{19}$$

where $\beta = (\alpha-\kappa)/2$.

6 Breakdown Boundaries

We probe the finite-sample limits of the SCD estimator by pushing the design toward extreme regions of the (n, K, T) parameter space. The goal is to identify configurations where nominal 95% coverage breaks down, either because the number of observations per group becomes too small to support reliable inference, or because the dimensionality of the donor pool overwhelms the available sample. We anchor $K \in \{5, 10, 25, 50\}$ donors, pair each with a short and long time horizon, and vary $n \in \{50, 200, 1000\}$. The data-generating process follows the simulation design of Rincón and Song [2026]. As a benchmark, we include the CS-DID estimator implemented in the `did` package (version 2.3.0; `https://github.com/bcallaway11/did`) of Callaway and Sant'Anna [2021], which is correctly specified under Scenario A but not under Scenario B. Results are based on $R = 1{,}000$ Monte Carlo iterations. Two scenarios are studied:

- **Scenario A**: True weights equal DID weights (uniform); λ = DID. CS-DID is correctly specified.
- **Scenario B**: True weights are sparse $(0, \ldots, 0, 0.1, 0.9)$; λ = Uniform. CS-DID is misspecified.

Entries marked "NA" indicate that SCD failed to converge because the number of observations per group is too small. CS-DID CI length under Scenario B is reported as "—" since CS-DID is misspecified and its CIs tighten around a biased estimate.

6.1 $K = 5$ Donors

Table 1: Coverage probability and CI length: $K = 5$, $T \in \{15, 55\}$, $n \in \{50, 200, 1000\}$.

		SCD Coverage		SCD CI Length		CS-DID Coverage		CS-DID CI Length	
n	T	A	B	A	B	A	B	A	B
50	15	0.996	0.984	5.634	3.750	0.972	0.631	4.147	—
	55	0.998	0.985	5.360	3.507	0.992	0.689	4.706	—
200	15	0.999	0.994	3.207	2.123	0.992	0.413	2.261	—
	55	0.999	0.996	2.969	1.964	0.995	0.451	2.540	—
1000	15	1.000	0.997	1.427	1.010	0.998	0.190	1.030	—
	55	1.000	0.995	1.292	0.924	0.999	0.220	1.166	—

6.2 $K = 10$ Donors

Table 2: Coverage probability and CI length: $K = 10$, $T \in \{20, 60\}$, $n \in \{50, 200, 1000\}$.

		SCD Coverage		SCD CI Length		CS-DID Coverage		CS-DID CI Length	
n	T	A	B	A	B	A	B	A	B
50	20	0.972	0.952	5.791	4.047	0.910	0.595	4.542	—
	60	0.972	0.950	5.408	3.788	0.941	0.690	5.199	—
200	20	1.000	0.995	4.140	2.779	0.993	0.471	2.778	—
	60	1.000	0.997	3.845	2.577	0.992	0.518	3.008	—
1000	20	1.000	0.999	1.940	1.327	0.994	0.236	1.293	—
	60	1.000	0.994	1.806	1.221	0.999	0.278	1.432	—

6.3 $K = 25$ Donors

Table 3: Coverage probability and CI length: $K = 25$, $T \in \{35, 75\}$, $n \in \{50, 200, 1000\}$.

		SCD Coverage		SCD CI Length		CS-DID Coverage		CS-DID CI Length	
n	T	A	B	A	B	A	B	A	B
50	35	NA	NA	NA	NA	0.786	0.526	4.327	—
	75	NA	NA	NA	NA	0.784	0.586	4.847	—
200	35	0.985	0.971	4.362	3.215	0.952	0.517	3.617	—
	75	0.982	0.972	3.630	2.836	0.958	0.571	3.938	—
1000	35	1.000	0.996	2.784	1.874	0.995	0.325	1.890	—
	75	1.000	0.998	2.629	1.710	0.998	0.343	2.052	—

6.4 $K = 50$ Donors

Table 4: Coverage probability and CI length: $K = 50$, $T \in \{60, 100\}$, $n \in \{50, 200, 1000\}$.

		SCD Coverage		SCD CI Length		CS-DID Coverage		CS-DID CI Length	
n	T	A	B	A	B	A	B	A	B
50	60	NA	NA	NA	NA	0.642	0.468	4.040	—
	100	NA	NA	NA	NA	0.695	0.491	4.317	—
200	60	0.986	0.923	3.469	1.904	0.863	0.574	4.517	—
	100	0.958	0.815	3.343	2.293	0.875	0.558	4.500	—
1000	60	1.000	0.991	2.625	1.842	0.990	0.447	2.571	—
	100	0.997	0.985	2.089	1.558	0.998	0.430	2.641	—

References

Brantly Callaway and Pedro H. C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.

Ratzanyel Rincón and Kyungchul Song. Causal inference with groupwise matching. Working paper, Vancouver School of Economics, University of British Columbia, 2026.